Total Printed Pages: 03

Roll No. 0108EC221025

MAB-101
Examination –Dec- 2022
B.Tech. I Sem : Common for all Branches
Linear Algebra and Calculus

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Write the statement of Maclaurin's theorem Theorem. 02

(b) Expand **logx** in power of **(x – 1)** by taylor's theorem. 02

(c) Find the first four terms in the expansion of **log(1 + sin x)** by Maclaurin's theorem 03

OR

Find the radius of curvature of the following curves at the points indicated against them 03
$\sqrt{x} + \sqrt{y} = \sqrt{a}$, (1/4, 1/4).

(d) Discuss the maxima and minima of the function **U = Sinx Siny Sin (x+y).** 05

OR

Find the radius of curuature at the Pamt "t" of the curve. 05
x = 3a cost – a cos3t, y = 3a sint- a sin3t

Q.2 (a) If $\mathbf{u = \sin^{-1}(x/y) + \tan^{-1}(y/x)}$ Show that $\mathbf{x\,\partial u/\partial x + y\,\partial u/\partial y = 0}$ 02

(b) State and prove Euler's theorem 02

(c) if $\mathbf{u = (x^2 + y^2 + z^2)^{-1/2}}$ Show that $\partial u/\partial x + \partial u/\partial y + \partial u/\partial z = -u$ 03

OR

If $\mathbf{x^x y^y z^z = c}$ then show that $\delta^2 z/\delta x \delta y = -(x \log ex)^{-1}$. 03

(d) (d) If **u = f(r),** Where $\mathbf{r^2 = x^2 + y^2}$ Show that $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 = f''(r) + 1/r\, f'(r)$. 05

OR

05

If $u = \sin^{-1}\left(\dfrac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}\right)^{1/2}$ then show that

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = \frac{\tan u}{144}(13 + \tan^2 u)$$

Q.3 (a) Evaluate

$$\int_0^2 \int_0^1 (x^2+y^2)\, dx\, dy$$

(b) Find the limit as $n\to\infty$ of the series

$$\frac{1}{n}+\frac{n^2}{(n+1)^2}+\frac{n^2}{(n+1)^3}\cdots\cdots+\frac{1}{8n}$$

(c) Find the area include between the parabola $y^2=4ax$ and $x^2=4ay$ 03

OR

Find the limit as $n\to\infty$ of the series 03

$$\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right)\right]^{\frac{1}{n}}$$

(d) Evaluate $\int_0^1 \int_0^x \int_0^{x+y} z\,dz\,dy\,dx$ 05

OR

Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\,dx\,dy$ and hence evaluate it. 05

Q.4 (a) Find the characteristic root's of the matrix $A=\begin{bmatrix}5 & 4\\ 1 & 2\end{bmatrix}$ 02

(b) Define Echelon form of a matrix with example 02

(c) Reduce the Matrix to normal form and find it's rank. 03

$$A=\begin{bmatrix}8 & 1 & 3 & 6\\ 0 & 3 & 2 & 2\\ -8 & 1 & -3 & 4\end{bmatrix}$$

OR

Verify Calley – Hamilton theorem for matrix. 03

$$A=\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\\ 3 & 5 & 6\end{bmatrix}$$



OR

Determine the eigen values and eigen vectors of the matrix 05

$$A=\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

5 (a) Define Equivalent, Contradiction. 02

(b) If p and q are two statements prove that $(p \wedge q) \Rightarrow p$ is a tautology. 02

(c) State and prove Demorgan;s law. 03

OR

(1) Degree of vertex (2) Walk (3) Isomorphic graph. 03

(d) Express the following function into disjunctive normal form: 05

$$F(x, y, z) = (x + y)(x + z')+(y + z')$$

OR

Simplify the following $(a+b).(a'+b').).(a'+b').=a'b'$ 05

$$pqr+ pqr'+ pq'r+ p'qr = pq + qr+ rp$$

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